

Question				Marking details	Marks Available							
					AO1	AO2	AO3	Total	Maths	Prac		
10	(a)	(i)			3			3				
				Neurotransmitter							Sympathetic Noradrenaline NOT adrenaline	Parasympathetic Acetylcholine (1)
				General effect							Excitatory/ stimulatory;	Inhibitory (1)
				Effect on heart rate							Increases	Decreases; (1)
1 mark for each row												
		(ii)		The larger the area on the motor cortex + increased movement {complexity/ increased fine movements} NOT number of motor neurones	1			1				
		(iii)		Increased sensory cortex to areas more receptive to stimuli / or example increased sensory cortex for genitals/ hand/ tongue etc (1)		1		1				
	(b)	(i)		Cortisol levels raised (1) Cortisol levels raised all the time (1) Negative Feedback loop {not working/ overridden/ faulty} (1)		2	1	3				
		(ii)		Epigenetic factors (1) Increased gene expression leading to increased cortisol levels (1)		2		2				
	(c)	(i)		Division of labour/ separate working groups/ different groups for different jobs/ 2 examples from: care of offspring/ looking for food/ reproduction/ defence of the colony (1) Increased efficiency (1)		2		2				
		(ii)		Pheromones/visual cues / touch/ movement		1		1				
		(iii)		Intra-sexual (selection)(1) Males more aggressive / increases sexual dimorphism / weaker males die off / strong males {survive/ reproduction}/ passing on of advantageous alleles (1)	1	1		2				

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		(iv)		Kinesis Accept: orthokinesis / klinokinesis	1			1		
		(v)		• Critical value of Chi-squared = 3.841 (1) • Calculated Chi-squared of 5.00 {> critical value/ to the right of the critical value} (1) • Reject {the null hypothesis/ H₀}(1) • there is less than a 0.05 probability the results are due to chance/ Significant difference between {the dry and humid environments/ O and E} (1)			4	4	2	3
				Question 10 total	6	9	5	20	2	2